Abhi Maddula

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EDUCATION

The University of Texas at Arlington May 2026
Bachelors of Science, Aerospace Engineering, Minor in Mechanical Engineering *GPA:3.2*
Awards: Presidential Scholarship, Augusta Johnson Memorial Endowed Scholarship
Organizations: AIAA, UTA Aeromavs, Student Advisory Board, Leadership Honors Program

SKILLS

CAD: SolidWorks, Siemens NX, CATIA V5 & 3DEXPERIENCE **| Software:** MATLAB, SimuLink, LabVIEW, ANSYS Fluent **|**
Manufacturing: 3D-Printing, CNC Milling (3-axis), Laser Metal Deposition (LMD-w) **| Programming:** Python, SQL **|**
Test & Hardware: High-pressure systems, Hydrostatic testing, Cold-flow testing

WORK EXPERIENCE

Bell Flight | Fort Worth, TX June 2025 – August 2025
Aerospace Engineering Intern (Bell MV-75 FLRAA)
- Performed **static load analysis** on **8+ engine interface** cases in **3DX**, ensuring compliance with engine specifications.
- Evaluated the **engine's drainage system** using **3DX** to calculate that the **combined flow rate** of the outlet is **less than 1%** of max discharge capacity, verifying compliance with engine fire zone drainage requirements.

GKN Aerospace | Fort Worth, TX February 2025 – May 2025
Mechanical Engineering Intern
- Led development of the **first data-driven performance tracking system** for laser wire DED operations, defining classification logic for deposition states and automating log file parsing using **Python and pandas**.
- Built an **automated analytics pipeline** using **Python** to generate runtime breakdowns and **improve manufacturing cell utilization by >5%**; results adopted across teams and **recognized by senior leadership**.
- Owned **end-to-end redesign** of post-manufactured scanning system, developing a **Siemens NX-based mobile cart** with optimized equipment layout, ergonomics, and storage; **increased scanning throughput and reduced per-part setup time**.
- Led a **cost and manufacturing trade study** on point cloud scanner evaluating vendor-supplied components vs. in-house fabrication, **minimizing total implementation cost** and reducing deployment timeline.
- Developed and integrated a **LabVIEW-based NI DAQ system** with **load cells and thermocouples** to capture **real-time thermal and force data**, focusing on **crack formation** during depositions in experimental builds.
- Identified need for protected instrumentation in a high-temperature welding environment and implemented corrective action by **designing a custom DAQ enclosure**, enabling reliable data collection and **validation of test results against simulation**.

PROPULSION EXPERIENCE

UTA AeroMavs – Hybrid Rocketry January 2026 – Present
Fluid Systems
- Led the mechanical design of a **liquid nitrous oxide feed manifold** and **pintle injector** for oxidizer operation.
- Developed a **Homogeneous Equilibrium Model (HEM)** in **MATLAB** to predict two-phase behavior of liquid nitrous oxide, calculating **theoretical injector sizing** for a target **mass flow of 0.112 kg/s at 750 psi**.
- **Mitigated machining risk** by targeting **Cd=0.8** to undersize the orifice, allowing **iterative testing** via empirical test data.
- Analyzed **injector performance** by comparing experimental data against **single phase incompressible** flow predictions to **evaluate discharge coefficients**.

Test & Integration
- Executed **hydrostatic proof testing** of nitrous oxide manifold and sealing system to **900 psi** using stepped pressurization, identifying **O-ring failure near 950 psi** and establishing safe operating limits.
- Conducted **cold-flow testing** of 3 injector configurations using **pressure transducers** and **load cells** to test mass flow rates.
- Supported **fluid system assembly and integration**, installing **JIC/ NPT** fittings, **ball valves**, **check valves**, plumbing, and instrumentation for **stable nitrous oxide cold-flow testing** and repeatability.

LEADERSHIP & EXTRACURRICULARS

American Institute of Aeronautics and Astronautics (AIAA) April 2025 – Present
Engineering Mentorship Director
- Leading UTA's first **mentorship program of 100+ students**, driving over **10 successful** interviews and job placements.

Hobbies & Interests: Rocket Propulsion**,** High-Speed Flight, Photography**,** Traveling, Weightlifting, Hiking, Volleyball, Pickleball